

$$\begin{aligned}
& c + di \\
& c + di + (4b + 9a)pd \\
& = (c + bpd) + di(4b + 9a) - dq(a + ib) \\
& = c + bpd - a^2d
\end{aligned}$$

INDIAN STATISTICAL INSTITUTE

SEMESTRAL EXAMINATION : (2024-2025)

M. MATH I ALGEBRA I

NOVEMBER 22, 2024 MAXIMUM MARKS : 50 DURATION : 3½ HOURS

By a ring we mean a commutative ring with 1.

Answer any two questions from 1, 2, 3, 4. [16 × 2]

- (1) (a) Let M be an R -module and I be an ideal of R . Prove that the R -modules M/IM and $M \otimes_R R/I$ are isomorphic. Deduce that $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/d\mathbb{Z}$, where d is the gcd of m, n (you may also give an independent proof). [6+5]

- (b) Let R be an integral domain and F its field of fractions. Is the R -module $F/R \otimes_R F/R$ the zero module? Justify your answer. [3]

- (c) Let N_1 and N_2 be submodules of R -modules M_1 and M_2 , respectively. Is $N_1 \otimes_R N_2$ a submodule of $M_1 \otimes_R M_2$? Justify your answer. [2]

- (2) Let M be an R -module. Define $\text{ann}_R(M) = \{r \in R \mid rm = 0 \text{ for all } m \in M\}$. We grant that $\text{ann}_R(M)$ is an ideal of R and $M/\text{ann}_R(M)M$ is naturally an $R/\text{ann}_R(M)$ -module. Also, the R -submodules of M are in one to one correspondence with the $R/\text{ann}_R(M)$ -submodules of M . Now answer the following questions.

- (a) If M is a Noetherian R -module, prove that $R/\text{ann}_R(M)$ is a Noetherian ring. [4]

- (b) Consider the $R[X]$ -module

$$M[X] := \{a_0 + a_1X + \cdots + a_nX^n \mid a_i \in M \forall i, n \geq 0\}.$$

Prove that

$$\text{ann}_{R[X]}(M[X]) = (\text{ann}_R(M))[X].$$

[4]

- (c) If M is a finitely generated R -module, show that $M[X]$ is a finitely generated $R[X]$ -module. [4]

- (d) Applying the above assertions, or otherwise, prove that if M is a Noetherian R -module, then $M[X]$ is also a Noetherian $R[X]$ -module. [4]

$$\begin{aligned}
& c + bpd - a^2d + di(4b + 9a) - dq(a + ib) \\
& = c + bpd + di(4b + 9a) - dq(a + ib) \\
& = c + bpd + di(4b + 9a) - aipd + aipd \\
& = c + di + pd(4b + 9a)
\end{aligned}$$

$$\begin{aligned}
& f(x, y) \\
& \text{as } f(x, y) = (1+x)^y
\end{aligned}$$

- (3) (a) Let R be a ring and M, N, P be R -modules. Prove that [8]

$$M \otimes_R (N \otimes_R P) \simeq M \otimes_R N \otimes_R P \simeq (M \otimes_R N) \otimes_R P.$$

- (b) Let R be a ring and let $0 \rightarrow N \xrightarrow{f} P \xrightarrow{g} M \rightarrow 0$ and $0 \rightarrow N' \xrightarrow{f'} P' \xrightarrow{g'} M \rightarrow 0$ be two exact sequences of R -modules, where P and P' are projective. Prove that [8]

$$P \oplus N' \simeq P' \oplus N.$$

- (4) (a) **Definition:** Let R be a ring and M, A, B be R -modules. Let $f : A \rightarrow M, g : B \rightarrow M$ be morphisms of R -modules. A fibre product of (f, g) is a triple (X, π_1, π_2) where X is an R -module, $\pi_1 : X \rightarrow A$ and $\pi_2 : X \rightarrow B$ are morphisms such that $f\pi_1 = g\pi_2$ with the following property: if (Y, p_1, p_2) is another such triple (meaning, there are morphisms $p_1 : Y \rightarrow A, p_2 : Y \rightarrow B$ such that $f p_1 = g p_2$), then there is a unique morphism $h : Y \rightarrow X$ so that $\pi_1 h = p_1$ and $\pi_2 h = p_2$.
Prove the existence of fibre product of (f, g) . [9]

- (b) Let I, J be ideals of R . Establish the following isomorphism of R -modules:

$$R/(I \cap J) \simeq \{(r + I, s + J) \in R/I \oplus R/J \mid r - s \in I + J\}$$

(No need to check that the RHS is an R -module). [7]

- (5) Answer any three questions [6 × 3]

- (a) Let $f, g \in \mathbb{Q}[X]$ be such that $fg \in \mathbb{Z}[X]$. Prove that the product of any coefficient of f with any coefficient of g is an integer.

- (b) Show that $XY - ZW$ is irreducible in $\mathbb{C}[X, Y, Z, W]$.

- (c) Let $a, b \in \mathbb{Z}$ be coprime. Prove that

$$\frac{\mathbb{Z}[i]}{\langle a + ib \rangle} \simeq \frac{\mathbb{Z}}{\langle a^2 + b^2 \rangle}.$$

- (d) Let $\varphi : \mathbb{R}[X, Y] \rightarrow \mathbb{R}[T]$ be the ring morphism defined by $\varphi(f(X, Y)) = f(T^2, T^3)$. Find the generating set of the kernel of φ .

- (e) Let R be a ring such that every nonzero ideal is a free R -module. Prove that R is a PID. (Hint: if a nonzero ideal has more than one generators, then the generating set cannot be linearly independent).

$$c + bpd = c + (abc + ia)pd + apdi$$

$$= c + apdi$$

$$= c + apdi + bpd$$

$$(-b + ia)pd + bpd$$

$$\begin{aligned} apc &= c - bpd \\ ap + bpd &= c \\ \varphi(ap) &= c - bpd + c + id \\ &= c - bpd + c + id \\ &= c + id + bpd + bpd \end{aligned}$$

$$-b + ia$$

$$(b + ia)pd + id$$

$$\begin{aligned} r - a &= s \\ r - b &= t \\ b + ia &= r - a \\ &= r - a + id \\ &= r - a + id + bpd + bpd \\ &= r - a + id + bpd + bpd \end{aligned}$$

$$\begin{aligned} (a + id, b + id) &= r - a + s \\ &= r - a + s \\ &= r - a + s \\ &= r - a + s \end{aligned}$$